Chapter 2

Technology and Engineering Potentials Needed

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2.1. Introduction

Identifying the core potential needed in technology and engineering is essential to succeed and to make a real impact. These potentials are like tools in a multidimensional toolbox: some sharpen your technical edge, while others shape the way you think and interact.

You will need strong analytical aptitude and excellent analytical skills, continually examine things and think of ways to help things work better. Attention to detail is important, the slightest error can cause an entire structure to fail. Review every detail thoroughly during the course of completing a project. You need to have excellent communication skills, translate complex technical lingo into plain English and communicate verbally with clients and other engineers working together on a project. Continuing your education, stay on top of developments in the industry, keep up-to-date of new research and ideas and be creative, think of new and innovative ways to develop new systems.

Other things you should be thinking of are making existing things work more efficiently, ability to think logically, making sense of complex systems and understand how things work and how problems arise.

2.2. Core Technical Potentials for engineering technology

These are the hard skills and knowledge areas that are foundational:

- Mathematical Thinking: Understanding calculus, algebra, and statistics for modeling and problem-solving
- Physics and Applied Sciences: Grasping mechanics, thermodynamics, and electronics to apply theory to real-world contexts
- Computer Literacy: Using software tools like CAD, MATLAB, and coding languages such as Python or C++
- Systems Thinking: Visualizing how parts work together in complex machines or infrastructure

- Design and Prototyping: Creating and testing tangible solutions with tools like 3D printing or simulation platforms

Excellent math skills, involve complex calculations of varying difficulty, good problem-solving skills. Frequently called upon solely to address problems, we must be able to figure out where the problem stems from and quickly develop a solution. Team players understand that one is a part of a larger team working together to make one project come together successfully and must work well as part of that team.

Excellent technical knowledge, you need to have an enormous amount of technical knowledge by continuous study and updating your knowledge. Understand a variety of computer programs and other systems that are commonly used during an engineering project.

2.3. Cognitive and Strategic Potentials

These powers shape your mindset as an engineer or technologist:

- Problem Solving: Breaking down challenges and approaching them with creativity and logic
- Analytical Reasoning: Interpreting data, troubleshooting, and drawing evidence-based conclusions
- Innovation and Curiosity: Exploring new technologies and asking questions like “what if…?”
- Project Planning: Budgeting, scheduling, and coordinating resources for maximum efficiency

2.4. Human and Communication Potentials

No system runs in isolation—neither do you:

- Teamwork & Collaboration: Working across disciplines and understanding diverse viewpoints

- Technical Communication: Explaining ideas through reports, diagrams, and presentations
- Ethical Reasoning: Thinking about long-term impacts, safety, sustainability, and equity
- Lifelong Learning Attitude: Staying updated with emerging tools like AI, green tech, and quantum computing.

2.5. Emerging Potentials for the Future

The engineering tech landscape is shifting. Here are forward-looking potentials worth developing:

- Automation & AI Fluency: Understanding how intelligent systems shape engineering decisions
- Green & Sustainable Tech Insight: Designing with climate and resource constraints in mind
- Cybersecurity Awareness: Securing devices, data, and infrastructure in connected systems
- Interdisciplinary Flexibility: Adapting knowledge across fields like biology, electronics, and data science.

2.6. Making the Transition from High School to College

You belong to an elite group of students now because you are studying to be an engineer, 5% of B.S. graduates in the United States are engineers. You will be taught to look at your surroundings differently, you will learn how to ask questions to find out:

- How things are made
- How things work
- How to improve things
- How to design something from scratch
- How to take an idea from paper to reality and actually build something

What you do for the next four or five years will affect you for the rest of your life, how successful and happy you are will depend primarily on you by taking responsibility for your learning. In college, most of your education takes place outside the class, and has good study habits.

You should be thinking of developing some new study habits that work and get rid of some old ones that don't.

2.7. Budgeting Your Time

Time is like money and very important in studying engineering or any other field, invest it wisely. Education is an expensive investment and requires responsible management. Learning is a lot of work, not much fun at the beginning, but it will become joyful, and your self-esteem will increase as you learn essential stuff. Learning and understanding new things can be downright exciting.

An example of weekly activities:

- Spend at least 2 to 3 hours of studying for each hour of class time. For example, if you are taking 12 credits, you should spend between 24 hours and 36 hours in studying time:

  (12 credits X 2 h = 24 h; 12credits x 3 h = 36 h)

Some classes will require more time for preparation and homework, projects, and lab work. Therefore, re-budgeting your time is important. Here is some tips to re-budget your time:

- Depending on how many hours a week you need to work and/or take care of family.
- Take breaks between classes or study.

Maintain a daily logbook to keep track of how closely you are following the schedule where time is being used inefficiently. Modify your schedule accordingly.

2.8. Study Habits and Strategies

You should always read the material before you go to the class and study the material after the class on the same day to get a full understanding of the topic you are studying.

Attend your classes regularly and get help right away. Take good notes, take notes when reading the textbook and listen carefully during the lectures. Study your notes for at least an hour or two the same day you take them, understand all the concepts and ideas that were discussed in class before you attempt to do your homework assignment. Here are other things you should do to learn the topics better:

- Review all examples in the chapter before doing the assigned homework problem
- Write notes in the textbook margins while studying
- Type a summary sheet of all important concepts and create an electronic file
- Keep all engineering books
- Select a good study place
- Study in a comfortable place with good lighting
- Avoid distractions
- Have roommates who understand your study needs
- Keep your desk clean
- Avoid having a picture of your sweetheart in front of you
- Form study groups
- Study on your own first, and then get together with your classmates to discuss and explain key ideas and concepts to each other.
- Everyone in your study group should agree to come prepared to discuss appropriate materials and contribute to the discussions
- Explain ideas and concepts in your own words to others in the group
- Be an active learner, don’t be a passive learner
- Prepare for examinations

There is no substitution for daily studying, therefore:

- Don’t wait until the night before the exam to study. The night before the exam is time for review only.
- Get a good night’s rest.
- Ask your instructor ahead of time what type of exam it will be, how many questions there will be and what suggestions she or he must help you prepare better. The best suggestions are above.

To reduce anxiety, prepare well and consider timing yourself when doing your homework problems.

2.9. Engineering Organization- get involved

Join an engineering organization! Networking, participating in design competitions, and obtaining student loans and scholarships are a few of the common benefits of belonging to an engineering organization.

ABET

ABET accredit college and university programs in the disciplines of applied science, computing, engineering, and engineering technology at the associate, bachelor, and master degree levels.

With ABET accreditation, students, employers, and the society, you can be confident that a program meets the quality standards that produce graduates prepared to enter a global workforce.

ABET began as the educational standard against which professional engineers in the United States were held for licensure. Today, after more than 80 years, ABET standards continue to play this fundamental role and have become the basis of quality for over 40 disciplines all over the world.

Developed by technical professionals from ABET’s member societies, our criteria focus on what students experience and learn. Sought worldwide, ABET’s voluntary peer-review process is highly respected because it adds critical value to academic

programs in the technical disciplines, where quality, precision, and safety are of the utmost importance.

ABET more than 2,200 experts come from industry, academia, and government. They give their time and effort supporting quality assurance activities around the world by serving as program evaluators, commissioners, board members, and advisors.

ABET is a nonprofit, non-governmental organization recognized by the Council for Higher Education Accreditation (CHEA).

Engineering Technology Department

The associate in science degree programs in Computer Engineering Technology, Electronic Engineering Technology, and Mechanical Engineering Technology are accredited by ETAC of ABET (Engineering Technology Accreditation Commission of ABET).

Attributes Model – ABET Engineering Criteria for Technology program CT, ET and MT at QCC 2025:

- Student Outcome (1) - an ability to apply knowledge, techniques, skills and modern tools of mathematics, science, engineering, and technology to solve well-defined engineering problems appropriate to the discipline;
- Student Outcome (2) - an ability to design solutions for well-defined technical problems and assist with engineering design of systems, components, or processes appropriate to the discipline;
- Student Outcome (3) -an ability to apply written, oral, and graphical communication in well-defined technical and non-technical environments; and an ability to identify and use appropriate technical literature;
- Student Outcome (4) - an ability to conduct standard tests, measurements, and experiments and to analyze and interpret the results;
- Student Outcome (5) - an ability to function effectively as a member of a technical team;

IEEE

The Institute of Electrical and Electronics Engineers (IEEE)

A professional association with its corporate office in New York City and its operations center in Piscataway, New Jersey. It was formed in 1963 from the amalgamation of the American Institute of Electrical Engineers and the Institute of Radio Engineers. Today, it is the world's largest association of technical professionals with more than 400,000 members in chapters around the world. Its objectives are the educational and technical advancement of electrical and electronic engineering, telecommunications, computer engineering and allied disciplines.

ASME

American Society of Mechanical Engineers (ASME)

Professional association that, in its own words, "promotes the art, science, and practice of multidisciplinary engineering and allied sciences around the globe"

via "continuing education, training and professional development, codes and standards, research, conferences and publications, government relations, and other forms of outreach.“

ASME is thus an engineering society, a standards organization, a research and development organization, a lobbying organization, a provider of training and education, and a nonprofit organization.

Founded as an engineering society focused on mechanical engineering in North America, ASME is today multidisciplinary and global.

AIAA

American Institute of Aeronautics and Astronautics (AIAA)

Professional society for the field of aerospace engineering.

- The AIAA was founded in 1963 from the merger of two earlier societies: the American Rocket Society (ARS), founded in 1930 as the American Interplanetary Society (AIS), and the Institute of the Aerospace Sciences (IAS), founded in 1932 as the Institute of the Aeronautical Sciences.
- AIAA standards for aeronautics have been incorporated into broader standards and public laws. These standards help transition innovative aeronautical ideas to everyday usage.

AEI

Architectural Engineering Institute (AEI)

Professional organization for architectural engineers. It is managed as a semi-autonomous division of the American Society of Civil Engineers (ASCE), like the Structural Engineering Institute.

ASCE and ASHRAE are the delegated societies for the architectural engineering's ABET accreditation process. However, AEI has successfully developed, with NCEES, the architectural engineering professional engineering registration examination which was first offered in 2003.

2.10. Graduation Plan from College

Know your school's admission process, admission to the university, admission to the college or school and/or admission to a specific engineering program. Other stuff you need to know and do. Some of the tips to keep in mind:

- Know the requirements for each level of admission.
- Meet with your academic advisor and plan your graduation.
- Understand the prerequisites for each class and in which semester a class typically is offered.
- A program flowchart will be quite useful

For those of you who currently are studying at a community college and planning to transfer to a university later, you should contact the university, learn about their engineering course transfer policies and requirements, and prepare your graduation plan accordingly. Other Considerations, doing volunteer work, Not only will you feel good about yourself, you'll also gain a sense of satisfaction and feel connected to your community.

Class activity 1

1. What is the difference between engineering and technology?
2. Create a list of activities to increase your understanding of engineering careers. Develop a plan for implementing three of these activities?
3. What field of engineering are you interested in and explain why do you want to be an engineer?
4. Have you ever had a job you didn't like? Describe the job. What didn't you like about it? If that job played any role in your subsequent decision to major in engineering, explain what that role was.
5. What potential do you have to assist in you becoming an engineer?

References

ChatGPT. (2025). *Need Potential in Engineering Technology*. Retrieved from chatGPT: https://chatgpt.com/